STATUTORY PURPOSE & JURISDICTION: In accordance with Regulation (EU) No 376/2014, Regulation (EU) 2018/1139, and ICAO Annex 13, the sole objective of this multi-source technical occurrence dossier is the prevention of future aviation accidents and incidents, and not to apportion blame or liability.

1. OCCURRENCE IDENTIFICATION & KEY PARAMETERS

CLASSIFICATION	DATE & TIME (UTC)	AERODROME LOCATION	SPECIFIC AIRSIDE AREA
Serious Incident	14 Oct 2024 • 14:22 UTC	Toulouse-Blagnac (LFBO)	Stand 14 / Taxiway Papa

COMMERCIAL AIRCRAFT INVOLVED

Airbus A320-214 • F-GKXJ

Operator: Air France (Flight AFR 294 to Paris-CDG)
Persons on Board: 6 Crew, 142 Passengers (148 POB)
Operating Engines: 2x CFM56-5B4/P (Engines in start sequence)

GROUND HANDLING EQUIPMENT & CREW

Pushback Tractor TUG-04 • Heavy Towbar

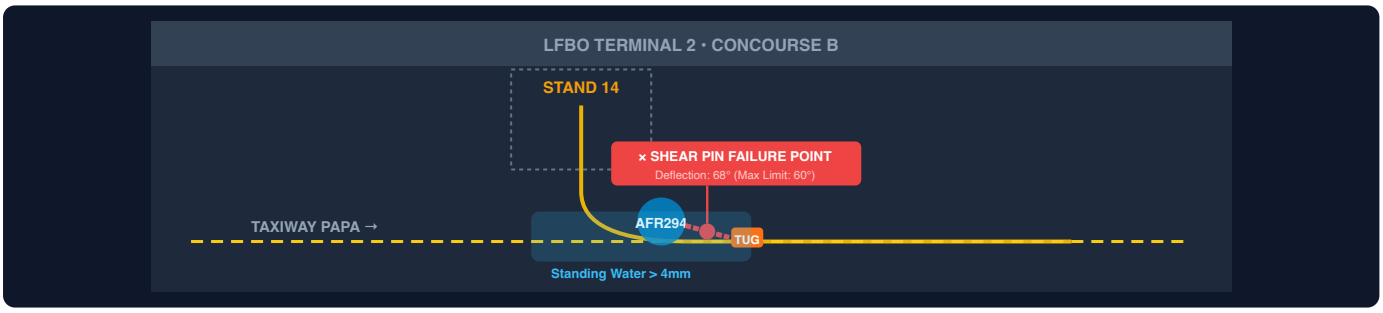
Ground Handler: Aviapartner LFBO Handling Services
Equipment Type: Schopf F396 Tractor + Mechanical Shear Pin Towbar
Ground Crew: 1 Tug Driver, 1 Headset Coordinator, 2 Wing-Walkers

2. EXECUTIVE CHRONOLOGICAL NARRATIVE

On 14 October 2024 at 14:18 UTC, Air France flight AFR 294 received pushback clearance from Stand 14 onto Taxiway Papa, facing South. The apron was contaminated by standing water following an active convective thunderstorm cell (+TSRA) with wind gusts recorded up to 38 knots. Simultaneously, Lufthansa flight DLH 104 was on final approach to Taxiway Papa via Taxiway Golf.

During the pushback pivot maneuver at an angle of 68 degrees, Blagnac Ground instructed Tug 04 to expedite movement to avoid blocking DLH 104. The tug operator applied compensatory steering input while braking on the water-slick apron. The mechanical shear pin within the towbar head fractured under excessive lateral shear force, releasing the towbar from the nose gear. The aircraft coasted under residual momentum before flight crew manual brake application arrested movement 4.2 meters from the taxiway center line.

3. APRON MOVEMENT SCHEMATIC & SHEAR POINT GEOMETRY



1. SYNCHRONIZED RADIO & INTERPHONE COMMUNICATIONS LOG

The following transcript is extracted from the Toulouse-Blagnac Ground Control voice recording system (Audio Channel 121.900 MHz) and synchronized with the Cockpit Voice Recorder (CVR) area microphone and flight deck interphone channels.

TIME (UTC)	STATION FROM	STATION TO	VERBATIM PHRASEOLOGY (SPOKEN ENGLISH)	OPERATIONAL NOTES
14:18:10	AFR 294	Blagnac Ground	"Blagnac Ground, Air France 294, Stand 14, Information Delta, ready for push and start, face South."	Normal Call
14:18:22	Blagnac Ground	AFR 294	"Air France 294, Blagnac Ground, push and start approved, face South, wind 280 degrees 22 knots gusting 36, QNH 1008."	Convective gusts reported
14:18:35	AFR 294	Headset (Interphone)	"Cockpit to Ground, push and start approved, facing South, release parking brake."	Interphone comms verified
14:18:42	Headset (Tug)	AFR 294	"Ground to Cockpit, steering pin in place, bypass valve confirmed, parking brake released, pushing back."	Towbar attached to NLG
14:20:15	Blagnac Ground	Tug 04 / AFR 294	"Air France 294, advise when tail is past the lead-out line on Taxiway Papa, Lufthansa 104 is vacating 14R via Golf."	Traffic sequencing notice
14:21:28	Blagnac Ground	AFR 294 / Tug	"Air France 294, if possible, expedite pushback on Papa, Lufthansa approaching your position."	Expedite Request
14:22:02	Headset (Tug)	AFR 294 (Interphone)	"Captain, we are swinging the tail South now, pavement is very slick, hang on."	Heavy rain noise on CVR
14:22:14	CVR Audio	Cockpit Area Mic	*[Loud metallic clack / mechanical snap followed by airframe shudder]*	Shear Pin Fractures
14:22:16	Headset (Tug)	AFR 294	"Stop, stop, stop! Set parking brake! Towbar has sheared! Brakes on!"	Emergency abort call
14:22:20	AFR 294 (Capt)	Cockpit / Ground	"Toe brakes applied, parking brake set! Aircraft stopped."	Aircraft stops on Papa
14:22:25	AFR 294	Blagnac Ground	"Blagnac Ground, Air France 294, stop pushback, our towbar has sheared, we are stopped across Taxiway Papa, tug is disconnected."	Airside obstruction alert
14:22:33	Blagnac Ground	DLH 104	"Lufthansa 104, hold position on Taxiway Golf, traffic disabled ahead on Papa."	Collision avoided
14:22:45	Blagnac Ground	Safety / Ops	"Ops 1, Blagnac Ground, Stand 14 / Taxiway Papa, towbar shear pin incident, dispatch airfield inspection."	Airfield intervention

2. ATC & HUMAN FACTORS ANALYSIS

PHRASEOLOGY & COMMUNICATION DISCIPLINE

- Standard ICAO phraseology was utilized during push clearance.
- The controller's prompt to *"expedite pushback"* introduced operational pacing pressure onto the ground team during an active turn in degraded surface conditions.
- High ambient precipitation noise impaired the clarity of the interphone link between tug driver and flight deck.

SITUATIONAL AWARENESS & SPATIAL CONSTRAINTS

- The convergence of DLH 104 created a bottleneck at the Papa/Golf intersection.
- The tug driver prioritized rapid clearance of the taxiway centerline over maintaining the maximum recommended 60-degree towbar pivot angle.
- Deceleration was initiated simultaneously with lateral steering input on wet asphalt.

1. AIRCRAFT TECHNICAL LOG & MINIMUM EQUIPMENT LIST (MEL) STATUS

TECH LOG EXTRACT • SHEET NO. 44108 • DATED 12 OCT 2024 — MEL ITEM DEFERRED • CAT C

MEL Reference:	**ATA 32-42-01 — Nose Wheel Steering (NWS) Control System**
Defect Description:	Intermittent 'WHEEL N/W STRG FAULT' recorded on ECAM during taxi roll at Paris-CDG. Tiller steering response sluggish below 10 knots.
Maintenance Rectification:	Troubleshooting deferred in accordance with MEL 32-42-01 Category C (10 calendar days). NWS electrical deactivation confirmed in unpressurized mode. Center pedestal placarded.
Operational Consequences:	During pushback, nose wheel castor free-rotation relies entirely on mechanical towbar bypass pin. Maximum permissible turn angle strictly capped at 60 degrees.
Authorizing Release:	Certified B1 Maintenance Engineer License #FR.66.014882 (Base CDG).

2. AERODROME METEOROLOGICAL OBSERVATIONS & RUNWAY SURFACE CONDITIONS

REPORT TYPE	TIME (UTC)	RAW OBSERVATION STRING	DECODED OPERATIONAL IMPACT
METAR	14:00 UTC	`LFBO 141400Z 28018KT 6000 –RA SCT012 BKN025 18/15 Q1012 TEMPO 4000 +SHRA=`	Moderate rain, declining visibility.
SPECI	14:20 UTC	`LFBO 141420Z 28024G38KT 1200 +TSRA FEW008CB BKN015 16/15 Q1008 WS ALL RWY TEMPO 28035G50KT +TSRAGR=`	**Severe Thunderstorm**, heavy rain, windshear, gusts 38–50 kt.
SNOWTAM	14:15 UTC	`LFBO 1014 14R 2/2/2 100/100/100 04/04/04 WET/WATER/WATER=`	Standing water depth 4mm on runway & apron taxiways.

WINDSHEAR & SURFACE FRICTION ANALYSIS

- Convective thunderstorm cell generated abrupt surface wind shifts (280° / 24 kt gusting 38 kt).
- Runway and taxiway friction coefficients fell below 0.35 due to standing water sheets exceeding 4mm.
- Hydrodynamic tire skidding risk was elevated for ground handling vehicles and aircraft gear.

TOWBAR MECHANICAL STRESS MECHANICS

- Shear pins are engineered to fail at 32 kN to prevent structural overload of the A320 nose landing gear trunnion.
- The measured angle of deflection at failure was **68 degrees**, exceeding the structural safety envelope.
- The failure worked as a design defense to protect airframe integrity.

1. CONFIDENTIAL MANDATORY OCCURRENCE REPORT (MOR) EXCERPTS

Captain Statement (AFR 294): *"We were initiating engine start on Engine 2 during pushback as permitted by SOP. Suddenly, during the tail pivot toward Taxiway Papa, we felt an acute airframe shudder and lateral jolt. Interphone communication was muffled by torrential rain pounding against the windshield. The ground coordinator called for immediate brake application. I applied manual toe brakes instantly and set the parking brake. The aircraft was disabled across Papa with the nose gear free-castoring."*

Pushback Tractor Driver Statement (Tug 04): *"I received instructions from Ground to expedite because an incoming A320 was vacating 14R. The apron was very wet with heavy water pooling near the Stand 14 drain. When I swung the tractor left to align the aircraft tail South, the tug wheels experienced momentary hydroplaning. I counter-steered while braking to regain heading, which caused an abrupt torque angle. Immediately I heard the shear pin crack. I called 'Stop!' on the headset right away."*

Ramp Operations Supervisor Statement: *"The ground handling team was working under severe turnaround time targets. A ground stop had been discussed but not yet implemented by the Airport Duty Manager despite the windshear alert. Both wing-walkers were in position but had severely degraded visual contact due to heavy rain sheets and spray from taxiing aircraft."*

2. SWISS CHEESE MODEL: LATENT TRAPS & ACTIVE DEFENSES BREAKDOWN

1. ORGANIZATIONAL

Latent Condition: Aggressive airline turnaround targets; lack of adverse weather pushback suspension thresholds in airport SMS.

2. ENVIRONMENTAL

Workplace Pre-conditions: +TSRA downpour, standing water >4mm, gusting crosswinds (38 kt), degraded acoustic headset clarity.

3. ACTIVE FAILURES

Unsafe Acts: Controller expedite instruction during turn; driver excessive turn angle (68°); combined braking and aggressive steering.

4. DEFENSES & MITIGATIONS

Barrier Status: Shear pin sacrificed to protect nose gear (worked); prompt abort call & toe braking prevented collision with DLH 104.

✓ INVESTIGATION DEBRIEFING PROMPT FOR GSEA ENGINEERS

In your investigation pair, synthesize these multi-source findings into a 2-minute oral presentation to the Airport Safety Directorate. Explicitly incorporate cause-and-effect connectors: *"owing to the standing water..."*, *"the shear pin failed as a consequence of..."*, and *"to mitigate this in future, the aerodrome operator must enforce..."*.